Human-centered design

Privacy/Behavior Analysis in Mixed Reality

- **Background**: Head-mounted mixed-reality devices can render and overlay realistic 3D content onto the physical environment, enabling a wealth of interactive possibilities. These devices are equipped with a rich set of sensors that collect personal and sensitive information (e.g., body motion, eye gaze, hand joints, and facial expression), which can be used to uniquely identify a user, even without explicit identifiers, and make inferences about human behavior
- **Goal**: Find useful and relevant datasets, preprocess them, and perform privacy/behavior analysis
- **Deliverables**:
  - Find existing datasets for user pose, gaze and gestures
  - Perform privacy/behavior analysis on the data based on existing literature
- **Software:** Python, Unity
- **References**:
  - MURMR: A Multimodal Sensing Framework for Automated Group Behavior Analysis in Mixed Reality, 2026
  - BehaVR: User Identification Based on VR Sensor Data, 2023
  - Understanding User Identification in Virtual Reality Through Behavioral Biometrics and the Effect of Body Normalization, 2021
  - Personal Identifiability of User Tracking Data During Observation of 360-degree VR Video, 2020
- **Team Size: 3**

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Project Motivation: VR systems collect sensor data including head and hand movement that have the potential to be used to identify users. Therefore, use of VR systems poses a possible privacy risk to users. We hope to determine how accurately a person can be identified from their VR data using a publicly available dataset and reproducible experiments. We have plenty of hardware experience both from the CompE core curriculum and from other classes such as 390B where Luke created an air quality sensing system and Aidan created a sun tracking solar charger in our respective teams. Furthermore, Aidan has experience with a data analysis project Aidan did sophomore year with the guidance of a grad student in Khwarizmi lab that has given me background knowledge in working with and processing large datasets in Python. Aidan's senior design project will also involve computer vision, user tracking, and gesture control which makes me especially interested in focusing on this project which will hopefully go hand in hand with my SDP.

Design Goals:

1. Measure whether users can be identified from head motion, controller motion, and eye gaze data
2. Identify which sensor signals contribute most to identification
3. Produce a reproducible analysis with clear charts, limitations, and recommendations for XR data collection

Deliverables: “Find existing datasets for user pose, gaze and gestures”, “Perform privacy/behavior analysis on the data based on existing literature”

System Blocks: Public dataset -> Preprocessing -> Feature Engineering -> Identification models -> Privacy analysis

- Preprocessing: Find users and sessions relevant dataset, handle missing values, and divide recordings into short time windows in preparation to process the data.
- Feature Engineering: Classify motion features such as speed and rotation, gaze features such as direction changes. Keep each sensor group separate for comparison.
- Identification models: Train a simple baseline classifier, then compare head motion, controllers, gaze and combined signals. Where the data allows, train on one session and test on a different session from the same users.
- Privacy analysis: Evaluate with what accuracy we were able to identify users based off of their data

Hardware and software requirements:

- Hardware: Two computers capable of processing CSV files; approximately 16GB Ram and sufficient storage for the selected recordings.
- Needed software: Python, Jupyter, pandas, NumPy, scikit-learn, and Matplotlib or Seaborn.
- Optional software: Unity to be used for a visual replay or demo if we have enough time.

Team responsibilities:

- 1: Dataset review, data cleaning, synchronization, feature extraction, and documentation of data limitations.
- 2: Identification models, privacy transformations metrics, and result visualizations.
- Both: We will choose the evaluation protocol, review each other's results, and write the report in accordance with our findings.

Timelines:

1. Review related material and confirm dataset access and license, define research questions
2. Inspect signals, users, sessions, and missing data; finalize the evaluation plan
3. Build preprocessing and time window feature pipeline
4. Train baseline identification models and compare sensor groups
6. Evaluate identification risk and create charts
7. Review results, limitations, and reproducibility, and draft the report and presentation
8. Finalize report, code, figures, and presentation

Expected deliverables: documented Python analysis, comparison charts, a privacy utility discussion and a final presentation